Civics Group Reg Number

Candidate Name ___________________________________

MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examinations 2011
Higher 1

H1 BIOLOGY **8875/2**

Paper 2

13 September 2011
2 Hours

Additional Materials: Writing Papers

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.
Write your name, civics group and register number in the spaces at the top of this page and all separate paper used.

<u>Section A (Structured Questions)</u>

Answer **all** questions in the spaces provided on the question paper.

<u>Section B (Essay)</u>

Answer **one** question on a separate writing paper provided.
Your answers must be in continuous prose, where appropriate.

At the end of the examination,
1. Fasten all answer papers for Section B securely.
2. Hand in **Sections A and B separately**

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [ ] at the end of each question or part question.

Examiner's Use	
Section A: Structured Questions	
1	/10
2	/13
3	/9
4	/8
Section B: Essay	
5 or 6	/20
Grand Total	/60

This paper consists of 8 printed pages

Section A: Structured Questions. Answer all questions (40 marks)

For Examiner's Use only

1(a) Describe how DNA is packaged in eukaryotic cells during interphase. [3]

(b) DNA replication occurs in cells during interphase before they divide by mitosis. Explain why it is important that replication occurs before mitosis. [3]

(c) State **two** differences in the behaviour of chromosomes in mitosis compared to meiosis. [2]

Mitosis	Meiosis
1.	
2.	

Some mothers of infants with Down's syndrome have a mutation of a gene coding for an enzyme which is involved in the metabolism of methyl groups. A lack of methyl groups on DNA results in mistakes in the segregation of chromosomes during cell division and in structural mutations of chromosomes. Individual with Down's syndrome has an extra chromosome 21.

(d) Explain how mistakes in segregation resulted in Down's syndrome. [2]

[Total: 10]

2. Fig. 2.1 and Fig. 2.2 are diagrams that show the main details of 2 processes.

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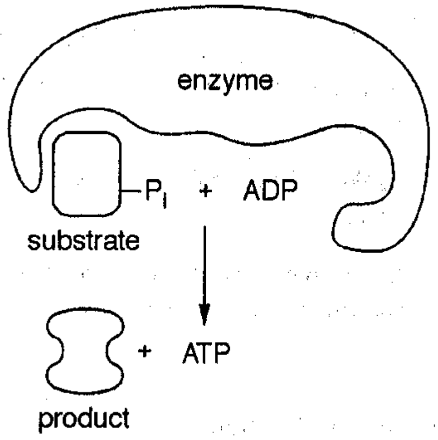

Fig. 2.1

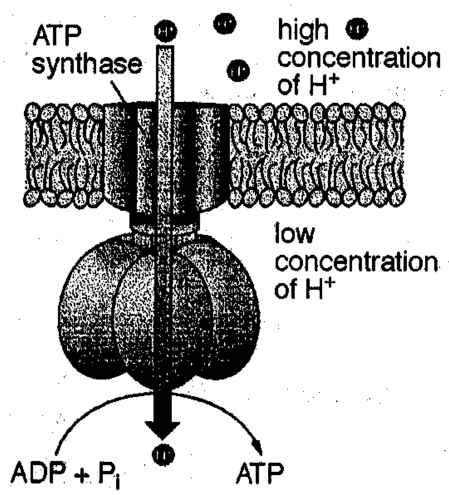

Fig. 2.2

(a) State precisely where these two processes occur in a cell. [2]

Substrate level phosphorylation :

Oxidative phosphorylation :

(b) Only substrate level phosphorylation is possible in the absence of oxygen.
Explain why oxidative phosphorylation is not possible in the absence of oxygen. [3]

For Examiner's Use only

Fig. 2.3 shows a protein complex found on the membrane of cells lining the stomach. An inhibitor acts by irreversibly blocking an enzyme in the protein complex. It is found that this inhibitor helps to reduce gastric acid secretion in the stomach.

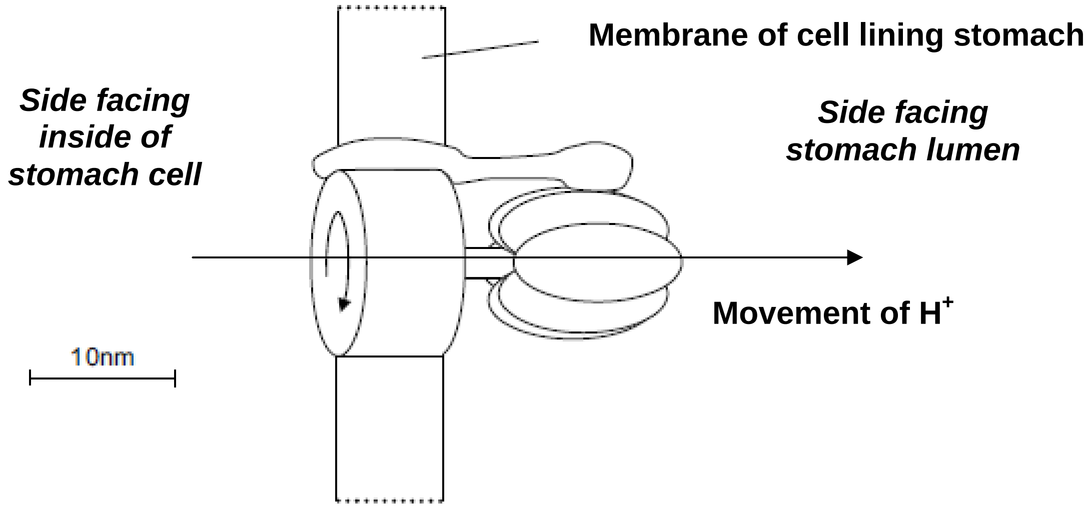

Fig.2.3

(c) Explain how the inhibitor helps to reduce the secretion of acid into the stomach. [2]

Researchers found out that these inhibitors could induce the apoptosis of gastric cancer cells.

(d)(i) Explain what is meant by the term 'cancer'. [3]

(ii) Name **two** factors that would increase the chances of the normal cancer gene becoming mutated. [2]

(iii) State one similarity between cancer cell and stem cell. [1]

[Total: 13]

For Examiner's Use only

3. Flax may be attacked by different races of a fungus, *Melamspora lini*, causing flax rust. One variety of flax, Ottawa, is resistant to race **24** of *Melamspora lini*, but is susceptible to race **22**. Another variety, Bombay, is susceptible to race **24** of *Melamspora lini*, but resistant to race **22**. When the two varieties of flax are interbred, the F_1 generation is resistant to both races of rust. One investigation into selfing the F_1 gave the following results:

128 plants resistant to both races **22** and **24** of rust,
39 plants resistant to race **22** but susceptible to race **24** of rust,
44 plants susceptible to race **22** but resistant to race **24** of rust,
14 plants susceptible to both races **22** and **24** of rust

(a) Using the given symbols, state, with reasons, the genotypes of the two parental varieties of flax. [4]

Symbols:
A - allele conferring reistance to race 22 *M. lini*
a - mutant allele; flax susceptible to race 22 *M.lini*
B - allele conferring resistance to race 24 *M.lini*
b - mutant allele; flax susceptible to race 24 *M.lini*

Genotypes: variety Ottawa : ______________________________

variety Bombay : ______________________________

Reasons:

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The allele for colour blindness, **g**, is recessive to the allele for normal colour vision, **G**. Complete colour blindness, unlike red-green colour blindness, is controlled by a different gene which is not on the X chromosome. The allele for normal vision, **B**, is dominant to the allele for complete colour blindness, **b**.

Fig. 3.1 shows the phenotypes of members of a family in which both types of colour blindness occur.

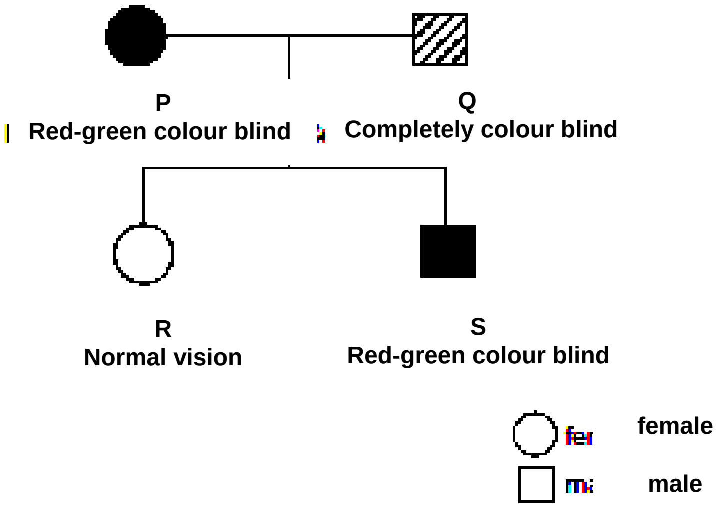

Fig. 3.1

(b) Using genetic diagrams and the symbols given, show clearly the cross between individuals **P** and **Q**. [4]

It was found that the complete colour blindness is due to a mutant allele on the chromosome. A normal allele codes for the development of normal cones (pigments cells in the retinal layer of the eye)

For Examiner's Use only

(c) Suggest how this mutant allele leads to abnormal development of cone cells. [1]

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[Total: 9]

4. The human insulin gene coding for insulin is inserted into a plasmid vector. Fig. 4.1 shows the plasmid vector. The polylinker contains 10 different restriction sites.

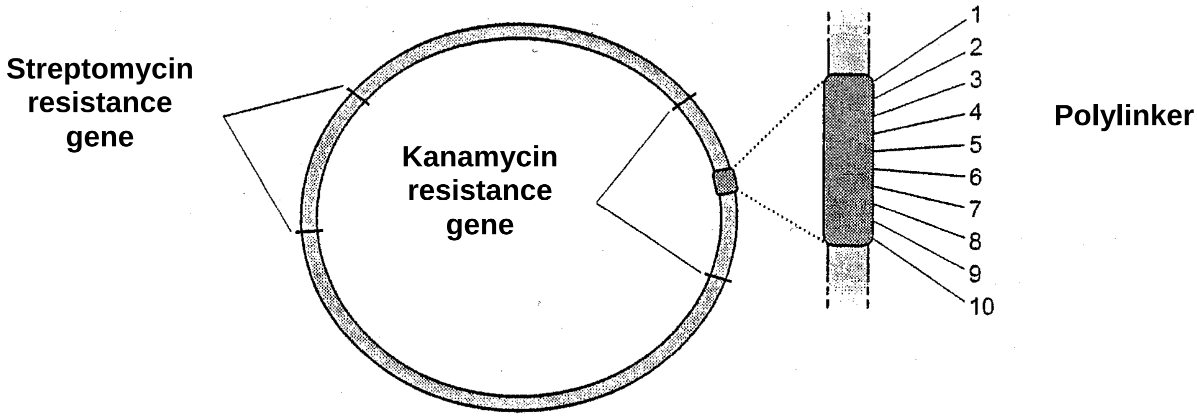

Fig. 4.1

(a) With reference to Fig. 4.1, explain the advantage of the plasmid having two antibiotic resistance genes and a polylinker within the kanamycin resistance gene. [3]

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(b) State a problem associated with cloning a eukaryotic gene into a prokaryotic plasmid vector and suggest a solution to overcome the problem. [2]

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(c) Discuss ethical issues raised by the use of genetically modified organism. [3]

[Total: 8]

Section B [20 marks]

Answer EITHER 5 OR 6.

Write your answer to this question on the separate answer paper provided.

Your answer:

- Should be illustrated by large, clearly labeled diagrams, where appropriate;
- Must be in continuous prose, where appropriate;
- Must be set out in sections **(a)**, **(b)** etc.., as indicated in the question.

Either

5 (a) Describe the main ways in which a globular protein differs from DNA. [7]

(b) Describe the role of messenger RNA in protein synthesis. [8]

(c) Discuss if DNA cloning can be considered as a form of evolution. [5]

OR

6 (a) Compare the structure and function of *Taq* DNA polymerase and RNA polymerase. [7]

(b) Outline how restriction enzymes are used for removing sections of DNA from a chromosome. [5]

(c) Explain how gel electrophoresis can be used to provide evidence of molecular homology. [8]

End of Paper